

Math 1314 Lesson 22

1) Use Synthetic Division to divide $\frac{-3x^5 + 5x^3 + 2x + 8}{x - 1}$

a) $\underline{\quad ? \quad} \mid -3 \quad \quad \underline{\quad ? \quad} \quad \underline{\quad ? \quad} \quad \underline{\quad ? \quad}$

b) $\frac{-3x^5 + 5x^3 + 2x + 8}{x - 1} = \underline{\quad ? \quad}$

2) Use Synthetic Division to divide $\frac{2x^5 + 4x^4 + 2x - 7}{x - 3}$

a) $\underline{\quad ? \quad} \mid 2 \quad \quad \underline{\quad ? \quad} \quad \underline{\quad ? \quad} \quad \underline{\quad ? \quad} \quad \underline{\quad ? \quad} \quad \underline{\quad ? \quad}$

b) $\frac{2x^5 + 4x^4 + 2x - 7}{x - 3} = \underline{\quad ? \quad}$

3) Use Synthetic Division to divide $\frac{-2x^4 + 3x^2 - 7}{x + 4}$

a) $\underline{\quad ? \quad} \mid -2 \quad \quad \underline{\quad ? \quad} \quad \underline{\quad ? \quad} \quad \underline{\quad ? \quad} \quad \underline{\quad ? \quad}$

b) $\frac{-2x^4 + 3x^2 - 7}{x + 4} = \underline{\quad ? \quad}$

4) Divide $-4x^4 + 7x^3 - 11x^2 + 2x + 9$ by $x - 2$

a) $\underline{\quad ? \quad} \overline{) -4 \quad \quad \quad \underline{\quad ? \quad} \quad \quad \underline{\quad ? \quad} \quad \quad \underline{\quad ? \quad} \quad \quad \underline{\quad ? \quad}}$

b)
$$\frac{-4x^4 + 7x^3 - 11x^2 + 2x + 9}{x - 2} = \underline{\quad ? \quad}$$

5) Divide $\frac{15x^2 + 12x + 8}{3x - 2}$

Hint:
$$\frac{15x^2 + 12x + 8}{3x - 2} = \frac{(1/3) \cdot 15x^2 + (1/3) \cdot 12x + (1/3) \cdot 8}{(1/3) \cdot 3x - (1/3) \cdot 2}$$
$$= \frac{5x^2 + 4x + 8/3}{x - 2/3}$$

a) $\underline{\quad ? \quad} \overline{) 5 \quad \quad \underline{\quad ? \quad} \quad \underline{\quad ? \quad}}$

b) $\frac{15x^2 + 12x + 8}{3x - 2} = \underline{\quad ? \quad}$

